



A Roadmap to Planning and Perform a ‘NMI – New Marketing Introduction’ for CAT Electric Drive Trucks

Maintenance and
Repair

Application

Component Life
Management

Component
Renewal (CRC)

MARC
Management

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December 2022
1.09-221212-1

1.0 Introduction

Developing an effective **NMI** (New Market Introduction) process is a challenging process for most CAT dealers. It is important that the Project Manager brings a 360-degree technical-operational vision of the Key Factors that are a part of the validation process for the product.

While there are many publications available on the **NMI** (New Market Introduction) and **NPI** (New Product Introduction), most tend to be more fragmented, highly specialized for a particular product, more maintenance focused or more operations focused. Typically, they also fail to connect and explore the integrated business vision with the customer's value chain, addressing aspect of **Total Cost of Ownership (TCO)**, **CPH (Cost Per Hour)** and **CPT (Cost Per Ton)**, **Maintenance Ratio (MR)** and more. With these it is not uncommon to have surprises once the trucks go into production cause a lot of rework or other financial intervention to achieve customer objectives for the machines.

This case study was an effective roadmap for introducing the Caterpillar electric-drive **794AC's** into Brazil, a market dominated by **Komatsu** electric drive **930E-4's**; until the **794AC's** were introduced, there were no better options for this customer. Utilizing this roadmap was critical to achieving all the customer critical KPI targets for the 6 x 794AC trucks. The client was very impressed with the whole NMI process (from preparation through implementation through actual operation/production). The high performance of this **794AC** fleet based on the successful NMI has given the customer confidence to move forward with additional 794AC purchases.

Other dealers facing challenges with NMI should be able to utilize the concept of this roadmap presented in this paper to successfully introduce new machines into their territory. SOTREQ is a big proponent of dealer collaboration and would be available to discuss this **NMI** in more detail.



Figure 1 – This roadmap supported the 794AC introductions in Brazil

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2.0 Best Practice Description

This Best Practice is a roadmap developed by SOTREQ for introducing the CAT **794AC** truck into Brazil with an approach well integrated into the mining customer's value chain. This same roadmap could be applied to **NMI** by other Dealers in other regions.

This introduction brings together the following:

- SOTREQ experience/expertise.
- Use of NMI documentation.
- Analysis of other operations (different models, types of customers/operations, etc).
- Interaction with Caterpillar **LMT/OHT** Product Groups.
- Specific product support and commercial (operations/customer) support at the site.

From concept standpoint, the 794AC NMI in Brazil was developed and build based on two key drivers: technical requirements needed, and commercial goals demanded. For this case, must consider these two ones:

Product Support Perspective – Key Technical Requirements

- Technical Mindset:** Utilize technical information already available from Caterpillar as well as other dealers...
- Strategy & Experience:** Build a **MSR (Maintenance Strategy Review)** and understand the risks with introducing the new model. Utilize the experience gained from other new model introductions at your site.
- Challenges - Learning & Assessment:** Utilize the lessons learned from other projects implemented on site: challenges of qualified labor, evaluation of customer performance (KPI's), indirect cost (CPH) that occur, etc.
- Strategy & Process:** Understand the business model for the operation / support of the new equipment on site - **MARC, CSA, CVA**, etc. Clearly establish the strategies and processes for parts, labor, tooling, shop productivity, etc.
- Customer Expectations:** Fully understand all the customer's **KPI** targets/required performance (Physical Availability, MTBF, MTTR, MR (Maintenance Ratio, Cost Per Hour, Cost Per Ton, etc.
- Put in place all the technical infrastructure in order to achieve the customer performance targets – safely and productively.

Operational & Customer Support – Commercial Goals demanded

- Benchmark the competition.** Understand what they are doing better or different than the Cat model on site and see how some might be applied to the Cat **NMI**.
- Consider innovation:** Technology, technical support, continuous improvement projects, safe maintenance, manpower productivity, etc.
- Understand your customer:** The Customer will always compare you with the competition, not only in relation to the quantitative indicators to be delivered but also to the qualitative aspects. Remember: The **NMI** Trucks are the “newcomer” to the site.

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- d. **Utilize the NMI platform:** Implement technical solutions that align with the customer's strategies and "business levers". Remember: The more you tailor to the customer's needs the better the chances that the customer will recognize the product as a success.
- e. **Create a New Communication Routine:** Create robust communication channels (daily, weekly, monthly, etc) to maintain visibility and partnership with the customer. Use these channels to demonstrate the differentiation between how Caterpillar/Dealer work with the customer compared to that of the competition.
- f. **Have a Marketing Plan:** Work with your marketing team to create a targeted plan that communication the right message to the customer.
- g. **Analyze Customer Insights:** Measure the opinion of the customer (mine management, operations and maintenance) with regular surveys to capture the thoughts and necessities of the customer. This will help you to adjust as frequently as necessary the processes in the project.

NPI CONTEXT

• 794AC MARKETING PLAN

- ✓ Mining Engineering worked for **4 months in 2020**, supporting the commercial team of mining machines and marketing in building the identity of the **794AC** truck for the Brazilian market.
- ✓ Construction of the CAT and Electric Diesel Truck technical identity (Not a Unit Rig);
- ✓ Construction of the CAT technical identity with a Payload of **320T** (Another option in the SOTREQ / CAT portfolio)

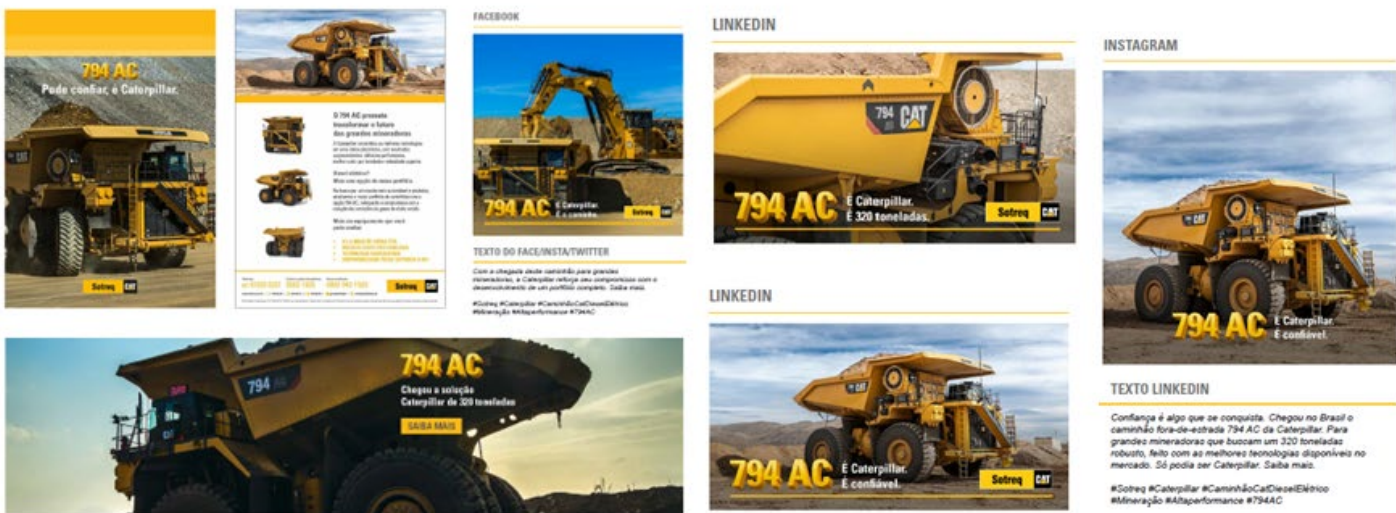


Figure 2 – Build a marketing plan to have customer confidence. Source: SOTREQ/Julio Correa

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3.0 Implementation Steps

Following are the key steps to implementing this roadmap/model for the introduction of a new model of truck or other machine into one's territory. It was successfully used for the **794AC** introduction into Brazil. It should be useful as other dealers introduce the **794AC** or any other new model into their regions.

Step 1. Risk Management / Maintenance Strategy Reference (MSR)

Step 2. Customer Value Chain

Step 3. Product Experience

- Parts
- Major Components
- Tooling
- Maintenance Strategy
- Component Management
- Training and Building People Knowledge

Step 5. Communication Plan

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ROADMAP NMI

INTRODUCING CAT ELECTRIC DRIVE TRUCKS

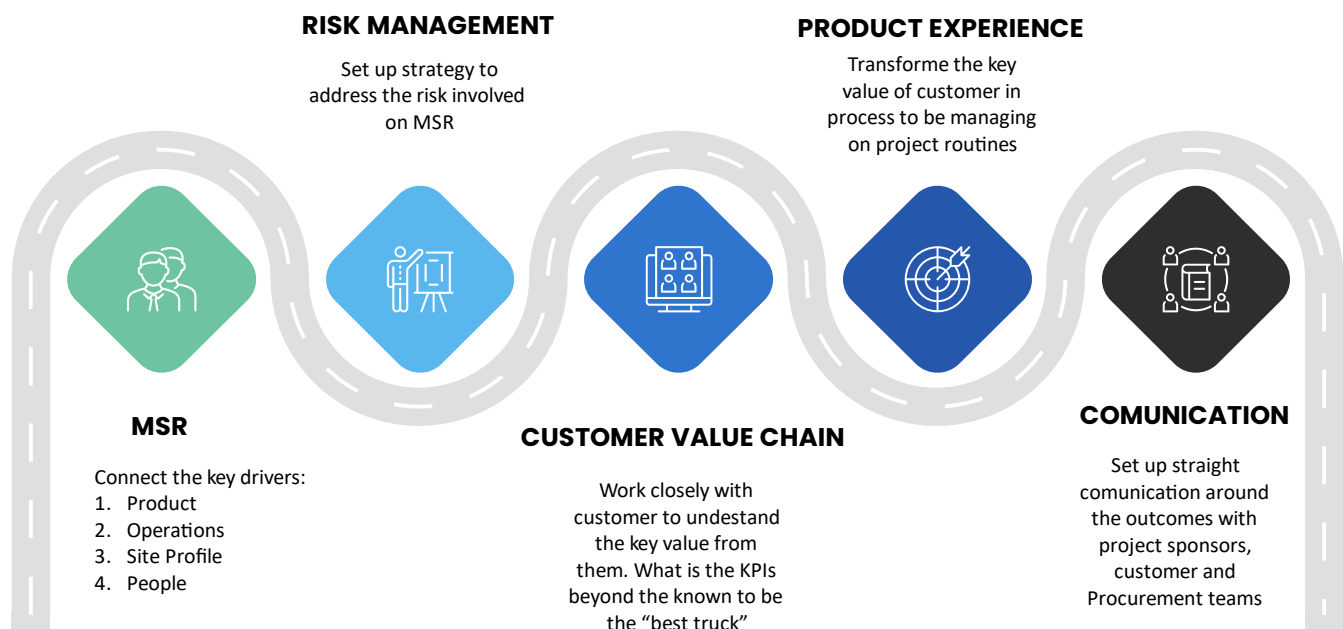


Figure 3 – The roadmap chart to dealer to set their **NMI** processes

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3.1. RISK MANAGEMENT / MAINTENANCE STRATEGY REFERENCE (MSR)

Understanding the risks of any new fleets is important to the success of any new fleet introduction. Capturing detailed technical information is the first step in constructing a risk based **MSR (Maintenance Strategy Reference)**.

A detailed Caterpillar Best Practice exists that can support you in your risk management efforts: ***Fleet Risk Management – Improving the 794AC Truck Performance Based on MSR***



Figure 4 – Best Practice published around the MSR and Risk Management as part of Roadmap

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3.2. CUSTOMER VALUE CHAIN

Doing an initial value survey with key site customer (in this case, maintenance, operations, and procurement) will help you to understand the important needs/acceptance criteria for the new product.

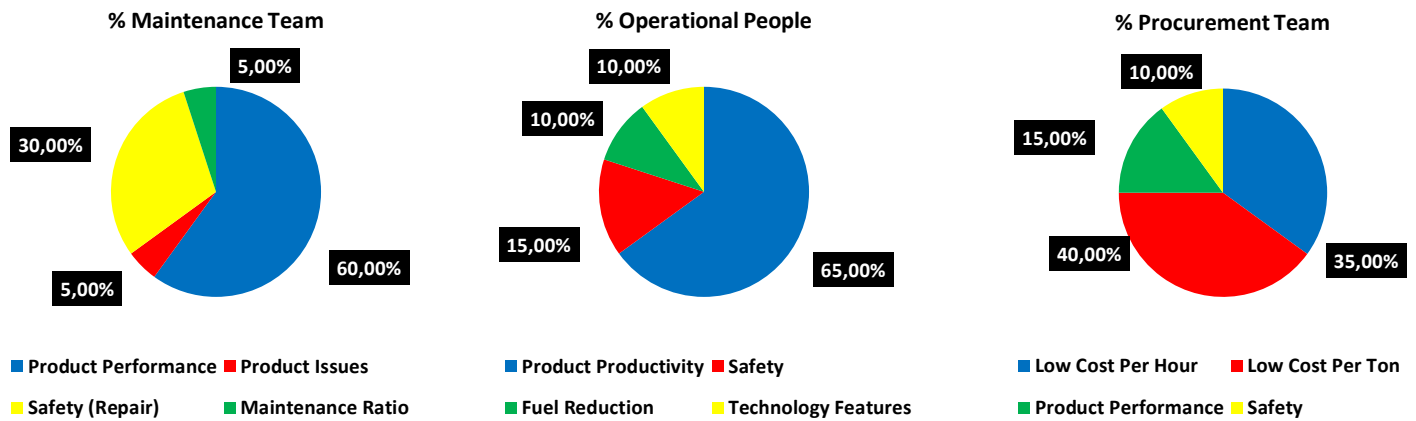


Figure 5 – **794AC NMI Survey**, Pre-work to build the Product Experience.

From the surveys, a risk / **MSR Matrix** can be developed which details the key drivers of the customer. With this, clear goals and objectives can be determined which fully align with the business vision of the introduction.

Below you can see the What, Why and How for the **794AC NMI** in Brazil:

What is important for customer?	Why? (VOC)	How to build it?
Parts service level > 90% (on site)	CAT parts shortage, PA > 90% / MTBF > 60 h, MTTR < 5 h.	Settled ABC stock strategy , dealer (B and C), CAT (A and B). 100% of items in Brazil.
Major mechanical components unplanned impact < 1% of PA	CPIs , Low reliability, unplanned failures other sites, part shortage, need to achieve performance KPIs	Some mechanical major components available on site
Major electrical components unplanned impact < 1% of PA	CPIs , Low reliability, unplanned failures other sites, part shortage, need to achieve performance KPIs	Some electrical major components available on site
No labor injures around 794AC maintenance repair. Safety improvement during maintenance repair	Comparison features between the major CAT competitor against 794AC	During machine configuration considered all safety devices and technology features available.
Reliable product – Product and Quality issues < 1% PA , MTBF < 10%	Bad reputation of 797F performance (Winning Together)	Have CPI parts on site, perform PIQ in assembly,
Maintenance Ratio and Parts Commonality with another CAT Models	Cost per hour., Cost per ton, Comparison with our major competitor	Perform Standard PM , address more improvement to reduce people on some repair process

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Low Fuel Burn Rate	High fuel cost thru mining site. Competitor has great performance in fuel burn	Address best fuel performance x productivity and find out fuel projects (CI) to optimize it.
Product Performance (People)	New introduction of truck in territory who major product is mechanical trucks.	Service training in Tucson (AZ) and develop an internal program to speed up the people training into 794AC
Product Performance (Tooling)	Former Product Introduction required a bunch of “fancy and expensive” tooling to perform maintenance	Build a Tooling list based on site needs not only based on RDTL suggested by CAT . Make a cross reference what competitor and CAT site trucks need, to show commonality between models.
Product Productivity	Have robust support to get more from truck to show its value	Regular field production study, Race Render® simulations, regular operational training.

3.3. PRODUCT EXPERIENCE

Once you have done your risk management assessment and understand the need/acceptance criteria, it is important to construct your **NMI** process focused on the product itself.

3.3.1. PARTS

A Parts Strategy for NMI needs to consider the service challenges, logistics, costs and required Service Level (i.e. >90%).

- **Building an Appropriate Spare Parts List:**

- Utilize the Caterpillar 794AC **RDSL (Recommended Dealer Stocking List)** as the starting point for the site parts list. Note: SOTREQ did a detailed review with the **794AC** Team where some adjustments were made to actual need and quantities.
- Utilize your backlog list and the parts consumption list (0 – 2000 hours) from other 794AC dealers around the world.
- Utilize your experience from other models on site (some which may have been captured through site **MSR's**)
- Understand parts related to current **CPI, PIQ** and other known failures from other dealers and adjust your list accordingly.

NOTE: SOTREQ'S **794AC** NMI Spare Parts list had 1,345 items.

- **Ensuring a Stocking Level of > 90%**

- Identify key service areas and group parts by those groups instead of a general “line item” list. Manage and take-action on parts by the groups. One such grouping is as follows: **A/C, Cab., Hydraulic, Electrical, PM, Minor Components etc.**

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- Define the list of NMI parts that will be stocked in country by the customer, dealer and by the Cat parts depot (servicing 100% of the **NMI** parts).
- Use the standard Dealer parts processes and CAT parts expediting process to manage the parts.

NOTE: Due to specific pressures of this **NMI**, monthly meeting between SOTREQ and Caterpillar (parts, engineering) were established to supplement the standard processes.

PARTS AND COMPONENTS MANAGEMENT

• TECHNICAL STRATEGY FOR CONSTRUCTION OF THE PARTS LIST:

- ✓ Loss profile analysis, backlogs from 0 to 20 thousand hours of all sites with 794AC;
- ✓ RDLS – Parts List recommended by CAT;
- ✓ Parts workshop with Product Group and Morton for electric trucks;
- ✓ Severity condition faced by other CAT models at the Carajas mine;
- ✓ Analysis of all CPIs, ICAs, ERFs, design changes;
- ✓ Monthly review based on the dealers' parts feedback network;



• OPERATIONAL STRATEGY FOR REPLACEMENT AND INVENTORY MANAGEMENT:

- ✓ Division of parts list by groups and with availability above 95%;
- ✓ Replacement on a daily basis over consumption;
- ✓ CAT strategy for most items in Piracicaba, not Miami;
- ✓ Direct channel with CAT team from LCC of PG for new additions in Piracicaba;
- ✓ Great commitment to CAT and IP Parts at the service level;
- ✓ Monthly meetings for inventory analysis and adjustment;



• COMPETITIVE ANALYSIS OF THE PARTS LIST 794AC X 930E (TCO / LCC):

- ✓ We compared the purchase prices of the main PM items, components, cylinders;
- ✓ Components and parts purchased with SEED / OTO with an average discount of 20% of CAT;
- ✓ Credits and AL obtained will enter the contract cost center;



Picture 6 – Parts Management developed into NMI **794AC** introduction

• Reviewing Parts Costs of the Competition vs. Cat

- Understanding the competition in terms of both maintenance and operations/productivity is important. Through direct numbers, discussions with management and technicians and observations, a lot of information about parts quality of parts (parts issues, component changeouts, etc) and costs can be gathered and factored into the **NMI** plan.

NOTE: In this **NMI** introduction, direct numbers for the competition were available. SOTREQ used this information establish parts benchmarks (customer always talking about low parts overall parts costs, etc) to best manage parts and components and show the best possible cost/hour for the **794AC** trucks.

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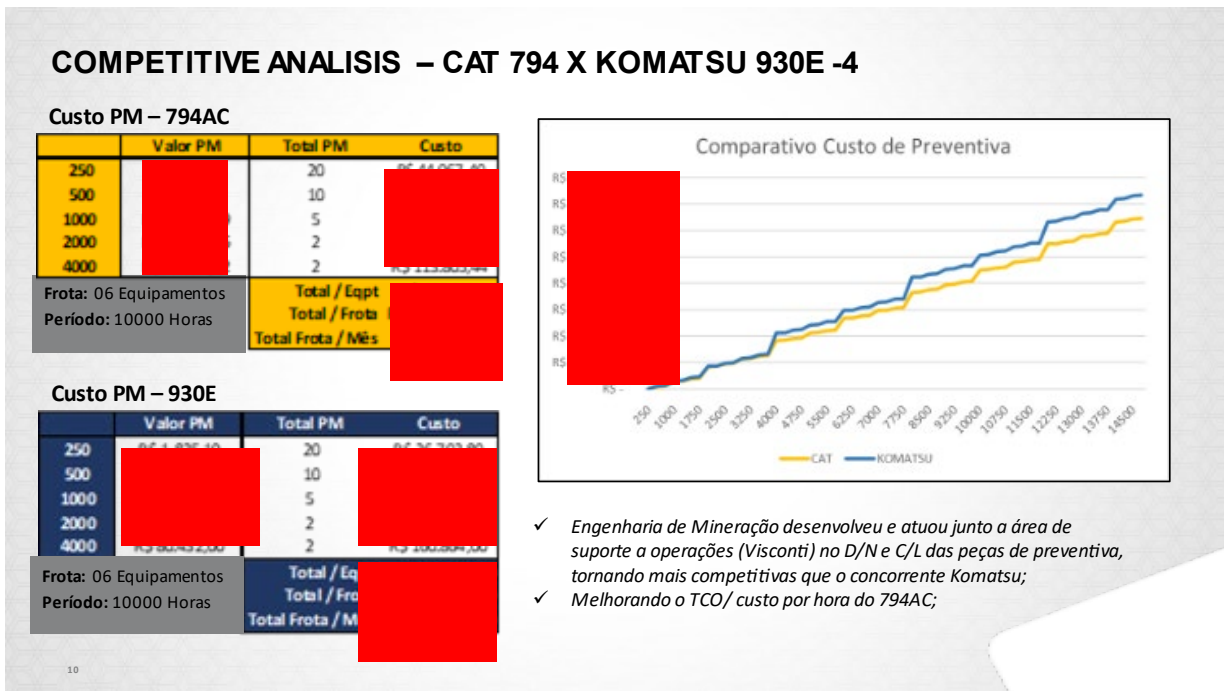


Figure 7 – Parts cost competitive analysis against the competitor Brazil site

3.3.2. MAJOR COMPONENTS

You will want to support the customer on establishing a plan for planned component removals as part of the NMI. Each customer has their strategies (hours, condition, condition + hours, budget availability, etc.).

SOTREQ developed a specific strategy for the **794AC** NMI which has the trucks on a two-year trial. A key feature of this strategy was that the impact of components needed to have **<1%** impact of Physical Availability. To reach this target and reduce capital, SOTREQ developed and applied the approach below. SOTREQ believe each dealer could “capitalize” on this approach to support their specific needs. SOTREQ would be happy to walk interested dealers through this approach.

1. Develop a Reliability Curve (Weibull Analysis) – starting with data available.
2. Evaluate Failure Rate of each component (CPI's/Warranty) vs. Impact on Physical Availability.
3. Account for the logistic availability challenges for critical parts/components.
4. Develop a Risk Buying Curve – gradual buying as the machines' SMU's increase.
5. Evaluation competition acquisition costs if the data is available and compare with Cat costs, determine best cost/hour/useful life ratio.

NOTE: For this NMI, 7 Major Components and 4 Hydraulic Components with high physical availability were purchased. See Figures 8 and 9 below for details.

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MAJOR COMPONENTS MANAGEMENT

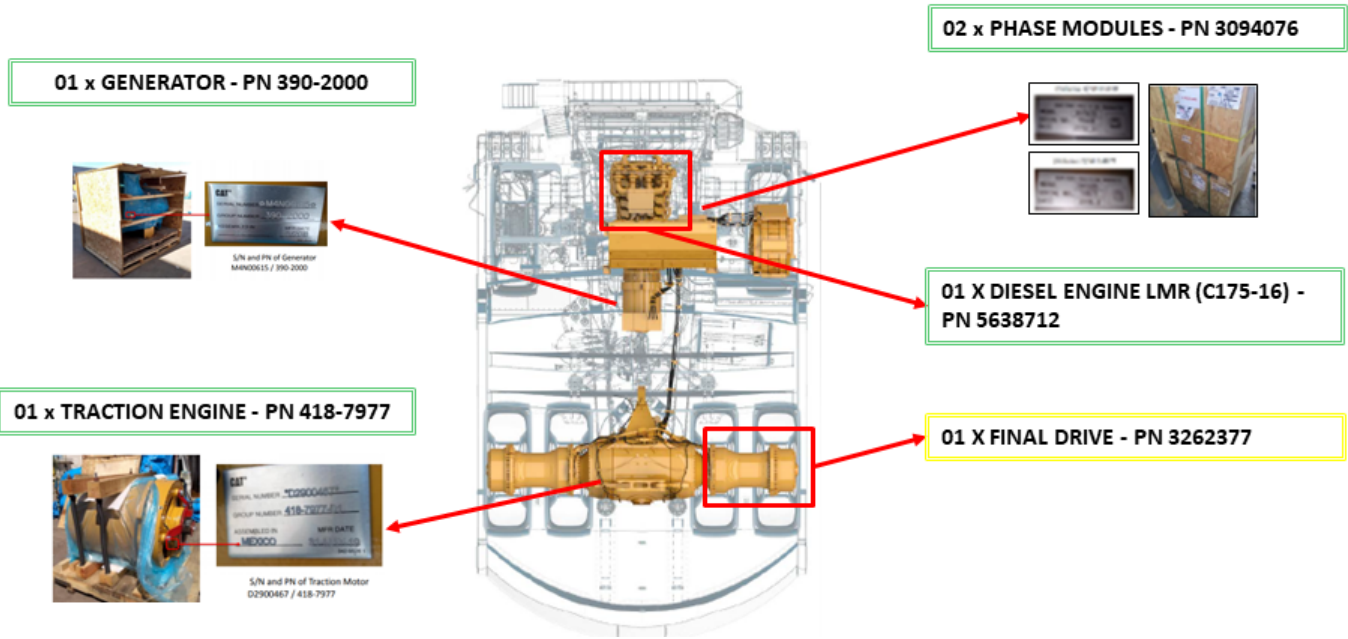


Figure 8 – Major components purchased during the **794AC** trial in Brazil

HYDRAULIC COMPONENTS MANAGEMENT

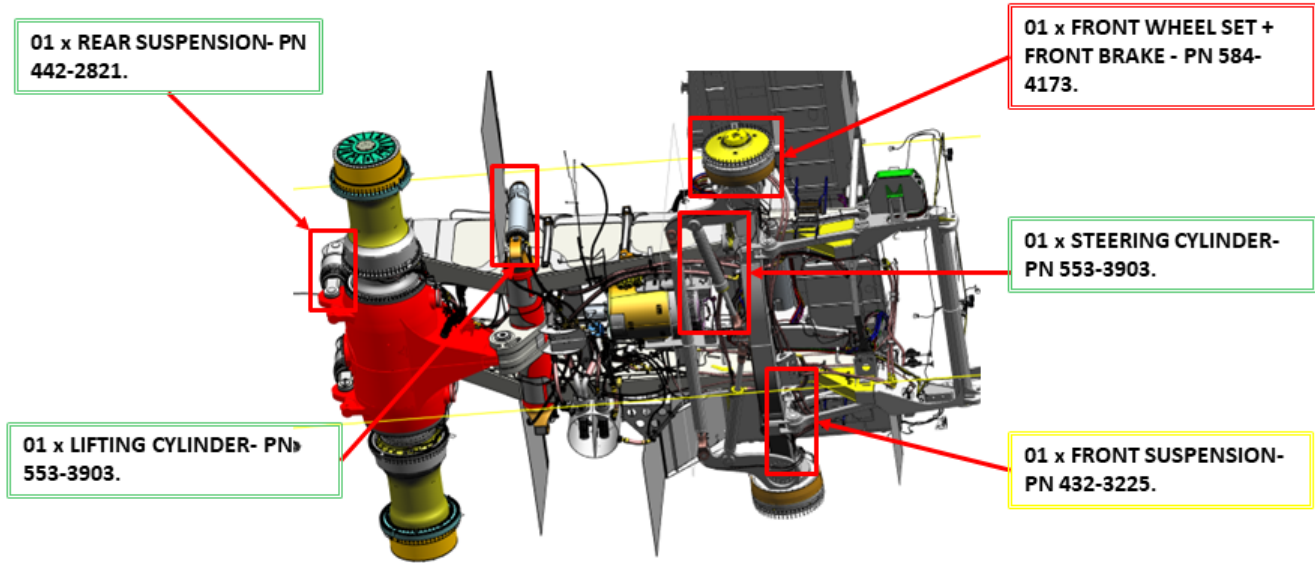


Figure 9 – Hydraulic components purchased during the 794 trials in Brazil

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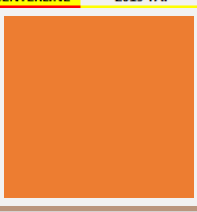
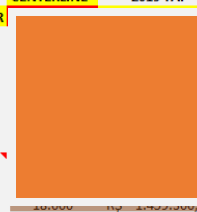
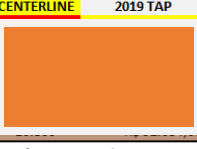
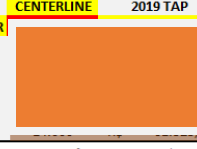
	930E-4	TCO Components			794AC	Dolar rate:	4,06	
		CENTERLINE*	2019 TAP			CENTERLINE*	2019 TAP	SID PROGRAM
AMONT =====>	PART NUMBER			AMONT =====>	PART NUMBER			4%
ENGINE CUMMINS				ENGINE 563-8712				
GENERATOR 58F-06-05170				GENERATOR 390-2000				
INVERTER MODULE(RGBT) 58F-43-00033				INVERTER MODULE(RGBT) 309-4076				
WM - TRACTION MOTOR				FINAL DRIVES				
(INCLUDED WHEEN DRIVE) 58F-30-00021				(INCLUDED REAR BRAKES)				
FRONT BRAKES GP 58F-32-40000				TRACTION MOTORS 418-7977				
BODY (AT SITE) 58F-MP-AK030				FRONT BRAKES GP 442-2571				
RADIATOR 58C-03-10120				RADIATOR 445-9770				
		BODY (FOB) 442-3125						
930E-4 Brazil (HYDRAULIC SYSTEM)				794 AC (HYDRAULIC SYSTEM)				
AMONT =====>	PART NUMBER	CENTERLINE	2019 TAP	AMONT =====>	PART NUMBER	CENTERLINE	2019 TAP	22%
CYL AS-SUSP (REAR) EM8846				CYL AS-SUSP (REAR) 442-2821				Product failures
CYL GP-HYD (HOIST) EM8356				CYL GP-HYD (HOIST) 469-5311				
SUSPENSION GP-A (FRONT) 58F-50-00101				SUSPENSION GP-A (FRONT) 432-3225				
BLOWER MOTOR/PUMP XA4146				DRIVE GP (PUMP) (PTO) 396-9741				
CYLINDER AS (STEERING) EM0241				CYLINDER AS (STEERING) 442-3555				
*5 years major components change outs performance of 830E at Carajás +				* Assumptions based on Major components changes outs performance towards				
Centerliner informed by Komatsu in 930E business deal with Vale (20% outliers off)				794AC sites, CAT specalog, and Carajás application (793,797)				

Figure 10 – Example: Cost per hour analysis, comparison cost x life between 794AC and competition

3.3.3. TOOLING

The added **NMI** Tooling requirements should be defined through a combination of the Tooling List provided by Caterpillar and the results of the **MSR** done and include the following:

1. Basic Truck Tooling not currently available on site.
2. **PM** Tooling, Diagnostic Tooling, Assembly Tooling currently not available on site.
3. Torque Tooling currently not available on site.
4. Tooling recommended by Caterpillar and other Dealer (based on their experience).

It is important to connect the tooling to the strategy of increasing productivity and safety of the workforce.

NOTE: In this NMI case, **SOTREQ** made numerous acquisitions that reduced assembly time and maintenance repair time while still maintaining a safety focus (**Elimination of Live Work**); the list is attached.

3.3.4. MAINTENANCE STRATEGY

The Maintenance Strategy must be built upon the MSR, Risk Management and those key aspects from the customer surveys. Discussions with other dealers on their experience (including downtime drivers) and strategies also are useful. Use of Best Practices, FPC Simulation and other tools will also support a great strategy.

In this NMI case, in addition to the above, there were additional KPI's as part of the commercial agreement that were also a foundation to the strategy.

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MAINTENANCE STRATEGY (MSR)

• HOW TO ACHIEVE UP 90% ON PA AND 60 HOURS MTBF

- ✓ Engineering made a great deepening in the profile of losses, CPIs, Performance of other dealers, Best Practices, FPC simulation in the last 6 months for the construction of the planned unavailability of the **794AC** contract;

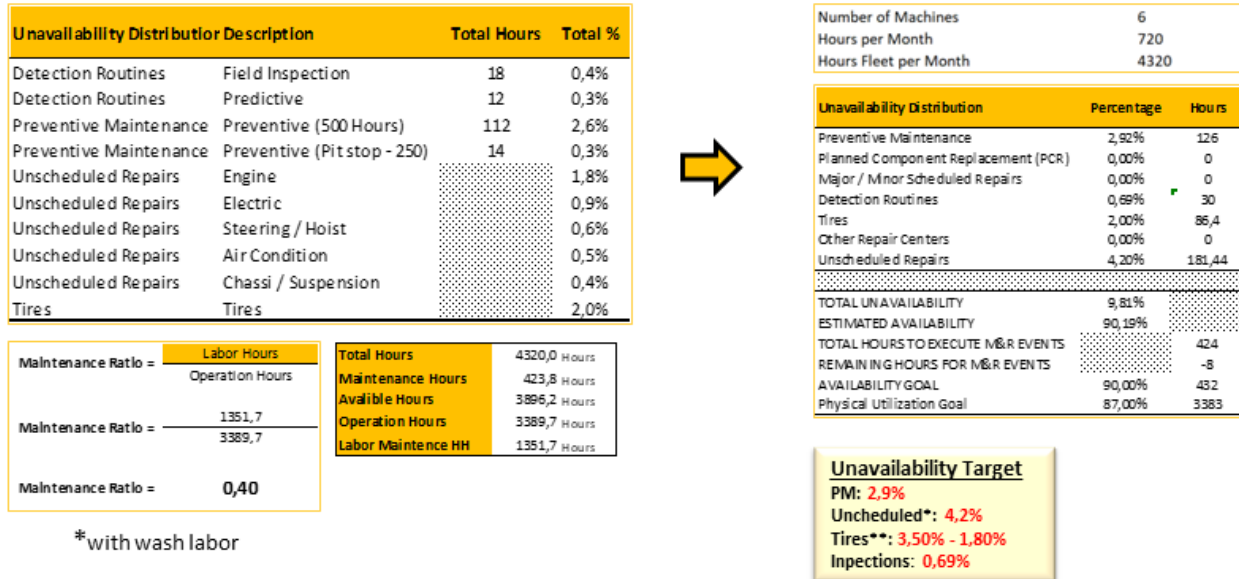


Figure 11 – Maintenance Strategy build to perform in high numbers of KPIs.

In this NMI, after defining the performance metrics anchored in the technical evaluation, SOTREQ developed a robust Maintenance & Repair Strategy and published this strategy as a Best Practice: **A PM Model to Deliver a Higher Availability for 794AC Trucks.**

 MINING DEALER BEST PRACTICE	
A PM Model to Deliver a Higher Availability for 794AC Trucks	
Maintenance and Repair Application Component Life Management Component Renewal (CRC) MARC Management	
A PM Model to Deliver a Higher Availability for 794AC Trucks	
1.0	Introduction
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8.0	Acknowledgements
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Figure 12 – Best Practice Published around the Maintenance Strategy thru this NMI Concept

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3.3.5. EQUIPMENT MANAGEMENT

Within the Product Experience section, it was emphasized that building a robust plan for all aspects of Equipment Management is essential for the customer to have all the benefit/feature of the product and a successful **NMI**.

Thinking Outside the Box

It is often necessary and important to think outside the box to find additional solutions to mitigate both operational and maintenance risks. Below are some of these “outside the box” solutions that **SOTREQ** has applied to this **NMI**.

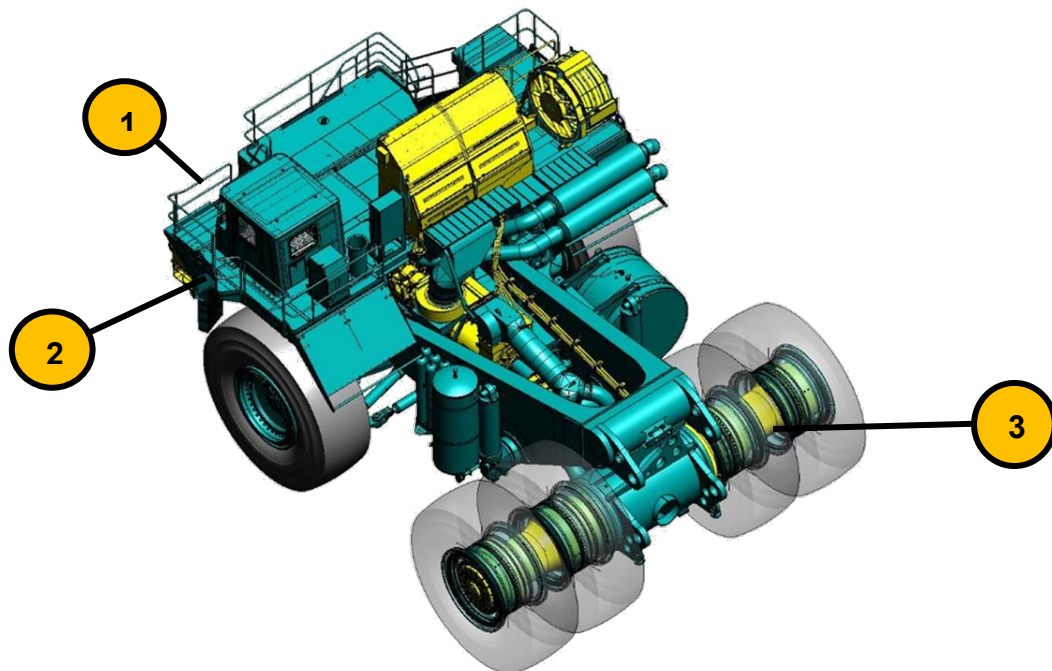


Figure 13 – Best Practice Published around the Maintenance Strategy thru this **NMI** Concept

CAT Products/Technology

- **1- Driver Safety System (DSS)** – DSS can be used to reduce the risk of accidents due to fatigue, typically at night. **(Safety)**.
- **2- Digital Video Recording (DVR)** – Used to access operator practices (good and bad) and their training. It has also been used as part of accident investigation. **(Safety and Productivity)**.
- **3 - Final Drive and Axle Oil 60 (FDAO 60)** – 60 FDAO (Synthetic Oil) is an option to extend the oil life and reduce cost/hour and maintenance ratio – all a part of the maintenance strategy **(TCO, Maintenance Ratio and Productivity)**.

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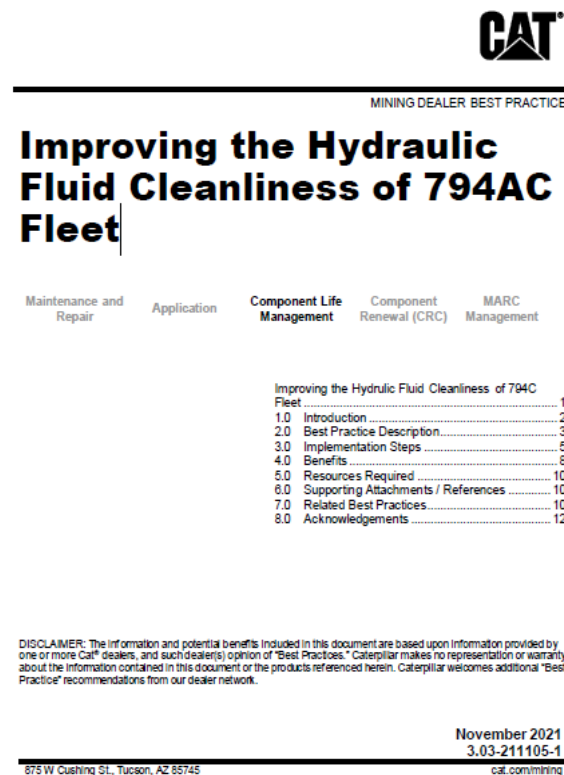
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Third-Party

- On-Board Kidney Looping can be one opportunity to increase productivity, reduce maintenance ratio and reduce cost/hour.

SOTREQ has developed and implemented a Caterpillar Best Practice that would be easily replicable to your truck fleets - ***Improving the Hydraulic Fluid Cleanliness of 794AC Fleet.***



Picture 13 – Best Practice Published around the kidney looping on board thru 794AC Trucks

Understanding the Mine Operation Model

It is important to understand the Mine Operation Model (**MARC, CVA, CSA**, other) for the site in building the Equipment Management Process. The customer is observing the “effort” that is spent to maintain the performance results (**Physical Availability, MTBF, MTTR**, etc.); a strong process will show that the **NMI** product is not a “heavy maintenance”! Bear in mind that most mining relates to **4.0 Industry** and are looking for mechanisms to reduce rework and manpower on site.

In this NMI, the operation model was a **CVA** with parts and labor under dealer responsibility and commitments in both performance and costs. SOTREQ took this opportunity to implement several mechanisms to support in this new mining environment.

NOTE: In the case of the typical mining dealer, they may look for market alternatives or simplified maintenance processes.

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MAINTENANCE SYSTEMS INTO 794AC TRIAL

- **QLIKVIEW & SOTREQLINK ON 794AC**
 - ✓ **SOTREQ** Group tools used as an official monitoring and information management platform for the **794AC**:
 - ✓ **SOTREQLINK & GFC:**
 - Platform for carrying out field inspections and PM.
 - Management of Condition Monitoring information.
 - Component Control.
 - ✓ **QLIKVIEW**
 - Management of all Maintenance KPIs (DF, MTBF, MTTR, SA, etc.)
 - Management of project costs and backlogs.
 - ✓ **ROADMAP OF OPERATIONAL ACTIVITY:** Mining Engineering mapped and created Guidelines for the 6 major processes of the **794AC** contract, describing the work routine of each employee



Figure 14 – M&R Processes Customized to Customer Site Managed by Web System

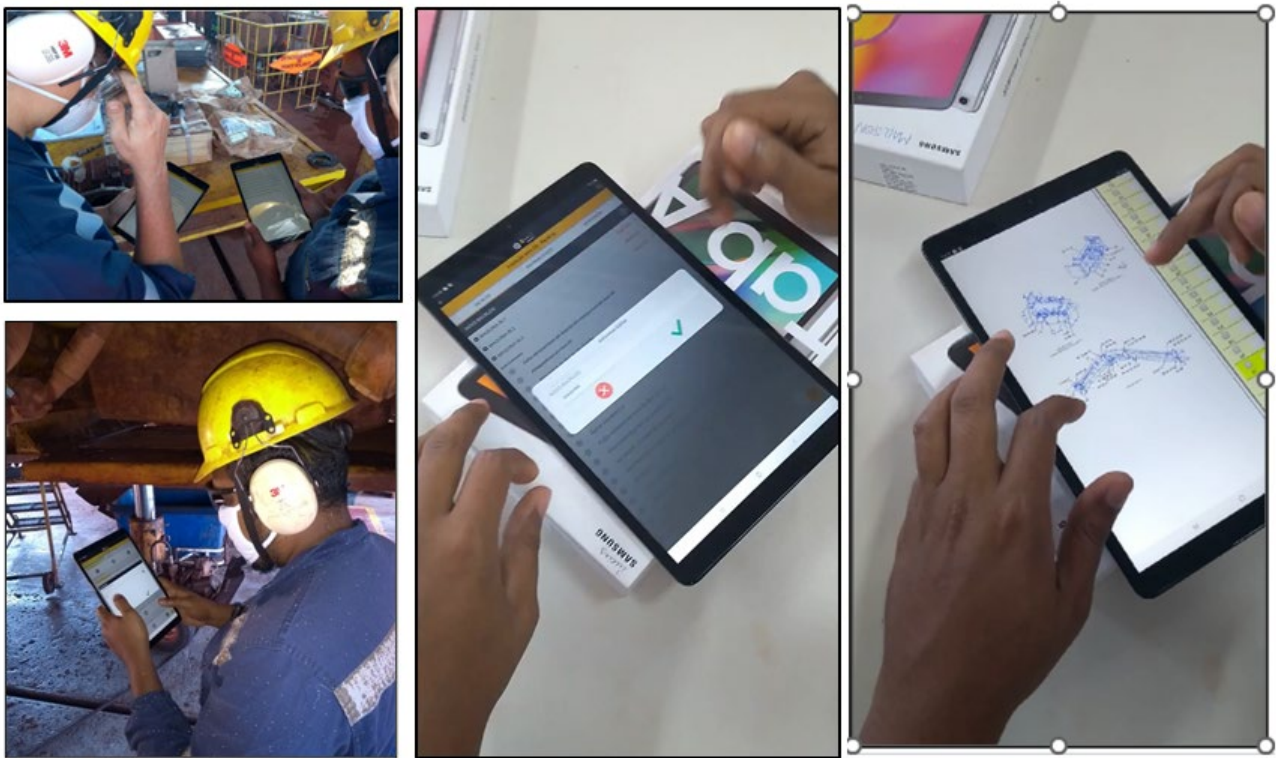


Figure 15 –Inspectors doing inspections with tablets, 25% - time improvement to have the backlog planned

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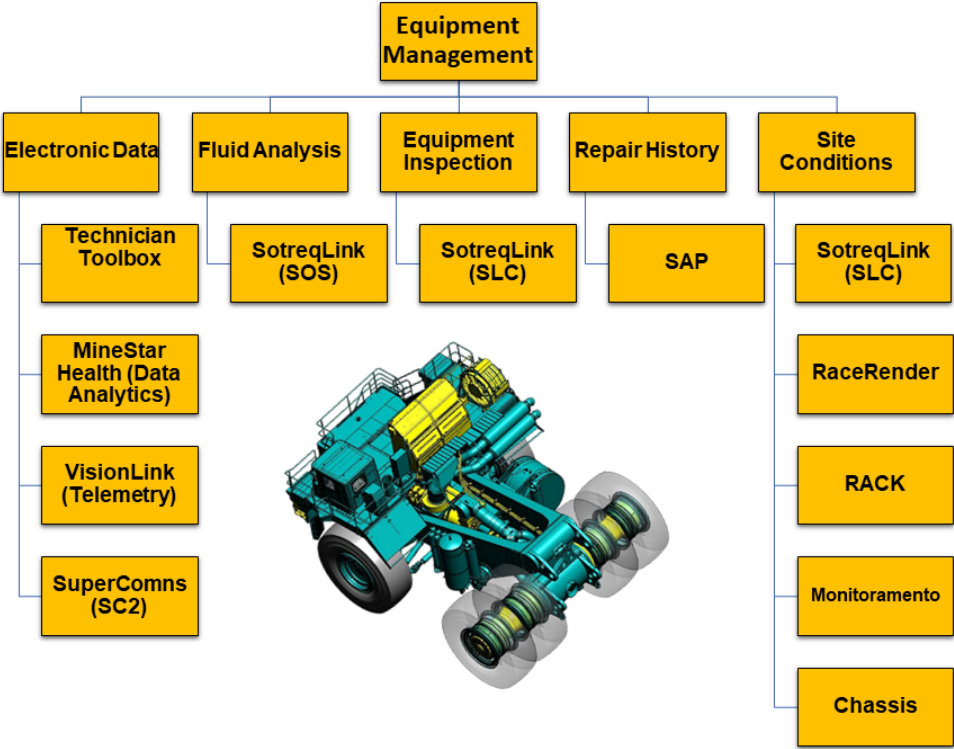


Figure 16 – Strong Condition Monitoring Strategy to address the risk found through MSR

As a reminder, use all the condition monitoring tools available (standard, new technology, etc) to capture and attend all product risks. Using **Supercomms** and **RaceRender** are two examples of using “New Technology”.

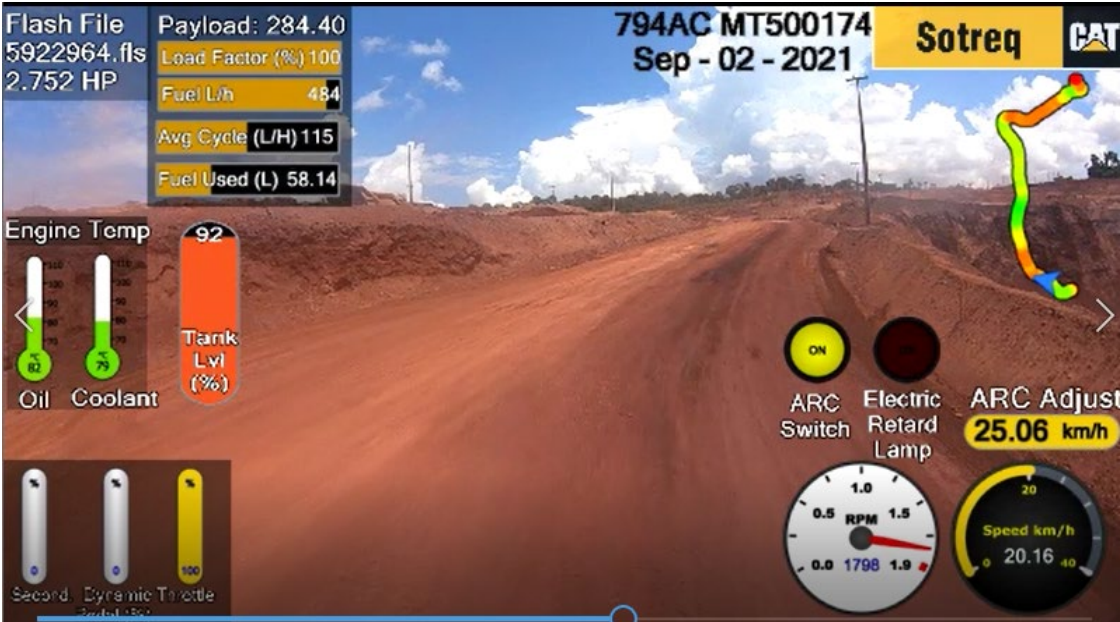


Figure 17 – Some software’s (Race Render) supported the condition monitoring strategy

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3.3.6. TRAINING PEOPLE & BUILDING KNOWLEDGE

Training and Building Knowledge are critical to the overall success of the **NMI** and future performance of the product. Many operations make the mistake of just looking for technical training associated with the product maintenance and the operation of the product.

The training and a knowledge base are necessities for everyone involved in the **NMI** process - from operators and mechanics in the field to site management to company management at corporate headquarters.

In this **NMI**, the engineers that developed the **NMI** process developed the training material; this detailed material was used to provide the technical and operations training in numerous training sessions throughout SOTREQ and the customer's organization to ensure the appropriate understanding and ultimately success of the **NMI**. To be more specific, over 320 hours of training was conducted throughout the **NMI** process.

Additionally, as there was a competitive product on site and there is constant comparison between the two products by the customer it was important to promote and defend our strengths. Adding the appropriate comparative information in the training material was successful used to by both the SOTREQ personnel and Customer to understand the positive differentiators of the Cat **794AC** over the competition.

SOTREQ developed "Playbooks" to support the training. If interested SOTREQ would be happy to share these playbooks.

TRAINING AND BUILD THE KNOWLEDGE

- **TECHNICAL SERVICE TRAINING**
 - People trained in **Tucson/AZ**.
 - **15 days** of interaction in the **United States**, including with Product Group.
 - **Workshop** at "The Cave", Development Team in **MOSHVILLE**.
 - More than **100 hours** of virtual workshops with **CAT**.
- **PROCESS AND NMI TRAINING**
 - ✓ **+ 120 hours** of training by the Mining Engineering team on Assembly to the Product Support team;
 - ✓ **+ 120 hours** of training by the Mining Engineering team on Preventive Maintenance to the Product Support team;
 - ✓ **+ 96 hours** of online training and **CAT** workshop for **SOTREQ** Product Support with Mining Engineering tutoring;
 - ✓ **+ 20 hours** of procedural training and **M&R** routines for **programmers** and **PM Supervisors**.
 - ✓ **+ 20 hours** of workshops on improvements, parts, components, processes, etc. with product support.
 - ✓ Negotiation to have a **CAT** operation instructor to train Customer
- **TECHNICAL WORKSHOP TO GET THE COMMITMENT**
 - ✓ Mining Engineering carried out several (10 people) surveys, evaluations, and technical indications over 6 months, of professionals (technicians) in the market who could be part of the operational staff of the **794AC** project.

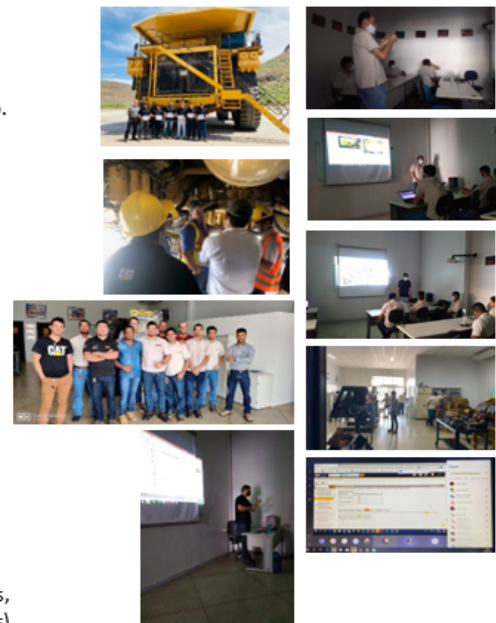


Figure 18 – Build the Knowledge around the **NMI** is one the most important part of product experience

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DOCUMENTATION AND TECHNICAL SUPPORT

- ✓ Mining Engineering worked for months to create a series of playbooks on **794AC** to support product support, focused on technical aspects of the product, maintenance operation and **TCO**.



- ✓ Additionally, material with technical comparison for the sales team and product support between the competitor **930E-4** and **794AC**. Serving as a basis for commercial and technical training at **SOTREQ**.



Figure 19 – Playbooks designed to training the shop people around the differences of CAT x Competitors

3.3.7. PRODUCT PROBLEMS MANAGEMENT

Depending on where your **NMI** machines were in the production line, not all updates and product improvements that are available to those machines may be on your machines. Consider implementing the updates and improvements during the assembly to minimize the impact of them in production.

Some key actions:

- Work with your Caterpillar **NMI** contacts to be fully aware of the updates and product improvements.
- Determine what should be fixed or applied during assembly.
- Plan appropriately for the required work (manpower, parts, scheduling, etc.).

Two important “windows of opportunity” for understanding and implementing fixes or quality corrections/updates are as follows:

- **Pre-Delivery Audit (PDA):** Request that PDA’s (“quality inspections”) are done on your trucks and review of those PDA’s. Confirm that all appropriate updates are made. If they cannot be corrected before delivery, make sure that you have the resources and planning in place to correct before commissioning.
- **Partners In Quality (PIQ):** Work with your Cat representatives to prepare for and perform all quality corrections and product improvements during machine assembly. Performing these corrections at machine assembly will minimize what needs to be done at PM or other

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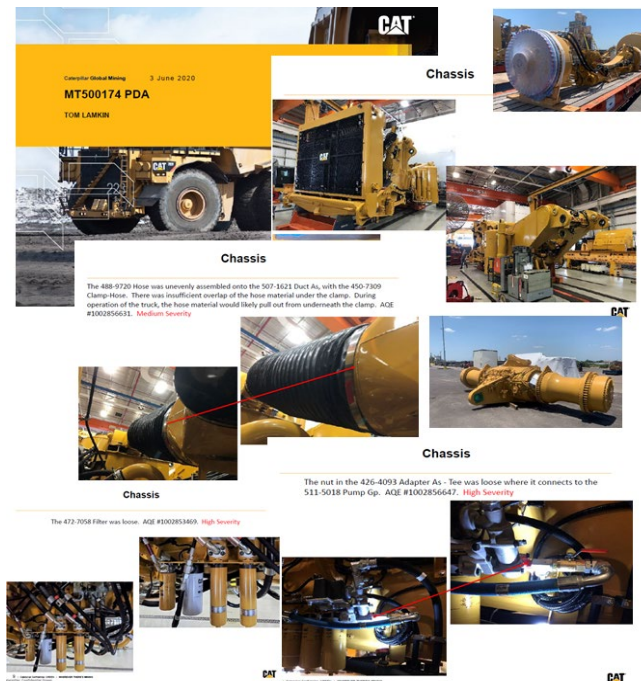
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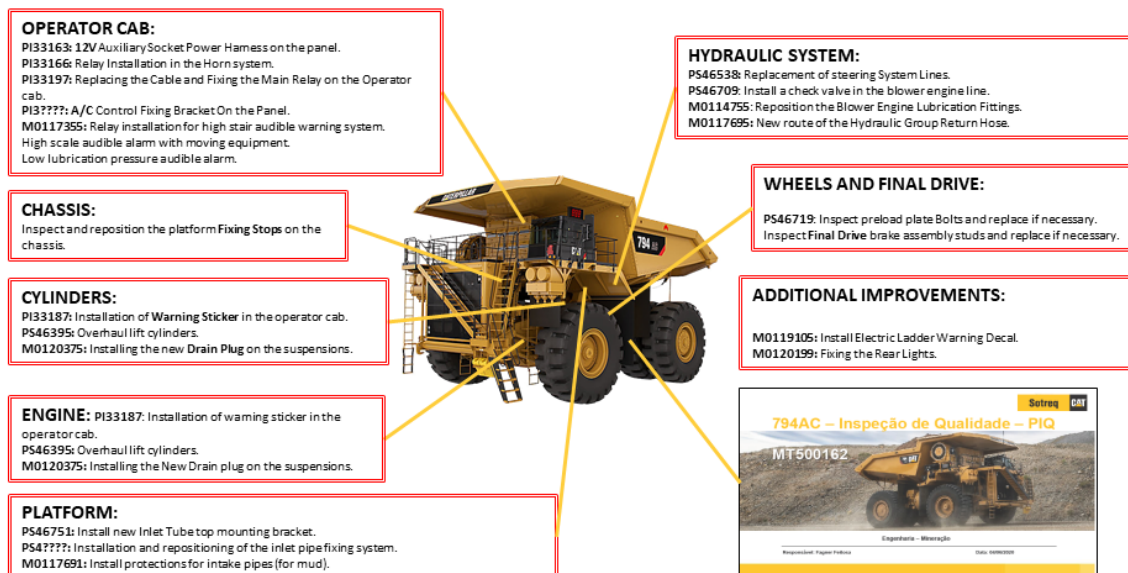
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downtime (scheduled or unscheduled) and support both better performance and better customer perception. It is important to alert the factory on the any issues found after the PDA.



Picture 20 – Make sure all trucks around the **NMI** made **PDA** before shipment.

PRODUCT UPDATES ON ASSEMBLY



Picture 21 – All the product updates made during the assembly

After NMI, standard technical processes should commence, including but not limited to the following: **PPM** Reviews, **DPPM** / **DSN** entries, Performance Reviews, CI Projects, technical rep visits, Parts Escalation, and more.

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3.3.8. COMMUNICATION PLAN

Building your **NMI** Plan in the project manager methodology of a **WBS – Work Breakdown Structure**, will give more organization and clarity on all the important aspects for introducing a new model into the region. Establishing a robust Communication Plan is critical for creating alignment and engagement throughout the process.

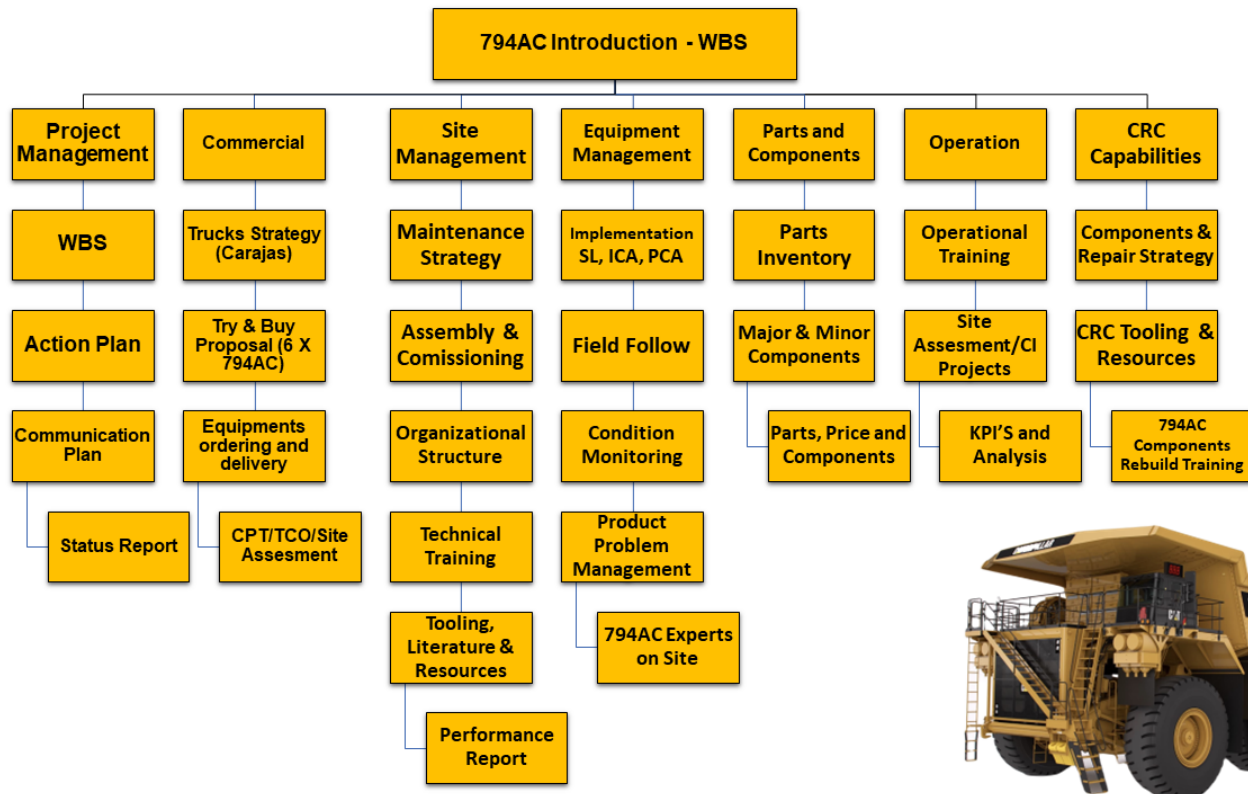


Figure 22 – Example of **WBS** build for **NMI** Process

Regular established meetings with proper participants, update status, actions, responsibilities are key to successful communication.

For this NMI, the following communication routine was established:

- **Week Meetings** (Follow Up Meeting) – Update the status of the **NMI** Plan with key process owners: Dealer Process Owners, PMO Team, CAT Region and NMI Teams.
- **Bimonthly Meetings** (Executive Meeting) – Asses the status of the implementation with stakeholders: CAT & Dealer Sponsors, Dealer Process Owners, PMO Team, CAT & Dealer NMI Team and Dealer Implementation Team.

NOTE: After **NMI** Implementation, it is suggested that a governance regime should remain in place for ~6 months after operations begins to monitor the performance and adjust as needed.

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4.0 Benefits

A few of the key benefit of this **NMI** Introduction, in addition to the benefits received through the individual best practices used in this document, are as follows:

- Better risk management through use of the **MSR**.
- Improved PM productivity by reducing waste.
- Improved contamination control and hydraulic oil life extension.

There are numerous results to demonstrate the success of this **NMI**. Some of the results are confidential but additional details may be shared through the SOTREQ management team. Here are some of the highlights:

- **Achieving KPI's for the 794AC Trucks:**
 - Physical Availability: > 90% for 1.5 years and going
 - MTBF: > 60 hours for 1.5 years and going
 - Cost Per Hour & Cost Per Ton: 5% less than the competition
- **Customer Satisfaction with the 794AC trucks, as determined through customer surveys:**
 - MR (Maintenance Ratio): 87%
 - Fleet Performance & Reliability: 93%
 - Productivity: > 90%
- **All of the CPH & CPT targets for the trial were met.**
- **The high performance of this 794AC fleet based on the successful NMI has given the customer confidence to move forward with additional 794AC purchases:**
 - 9 x 794AC Trucks at another Vale site.
 - 11 x 794AC Trolley Truck as part of the customer's carbon reduction focus for the future.
- **The customer recognized SOTREQ with award plaques for excellent fleet performance (PA, MTBF, MTTR).**



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5.0 Resources Required

We suggest that dealer who want to introduce their **NMI** with this roadmap have a **PMO** (Project Manager Office – Leader to embrace and connect all the people and resources required).

All the cost around the project – please contact the dealer to understand it.

6.0 Supporting Attachments / References

Contact the dealer to have access of all the parts and component list.

7.0 Related Best Practices

- *Fleet Risk Management – Improving the **794AC** Truck Performance Based on **MSR***
- *A **PM** Model to Deliver a Higher Availability for **794AC** Trucks*
- *Improving the Hydraulic Fluid Cleanliness of **794AC** Fleet*

8.0 Acknowledgements

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